

Phase 3 — Linux Hardening with Ansible

Harrington Capital plc | Infrastructure Engineer Simulation

Objective

Harden HC-APP01 and HC-MON01 against the Harrington Capital ISO 27001 security baseline using Ansible. No manual SSH configuration edits. Everything automated, everything committed.

By the end of this phase you will have: - SSH password authentication disabled on all Linux hosts - UFW firewall active with correct rules - fail2ban protecting SSH from brute-force - NTP synced to UK pool servers - Automatic security updates configured - Ansible PLAY RECAP as evidence

Prerequisites

- Ansible installed on your control machine (your laptop or HC-MON01):

```
sudo apt install -y ansible
ansible --version
```

- SSH key-based access from your control machine to HC-APP01 and HC-MON01:

```
# If not already set up:
ssh-copy-id hcadmin@<HC-APP01-IP>
ssh-copy-id hcadmin@<HC-MON01-IP>
```

- Inventory file updated with correct IPs: Edit `/ansible/inventory.ini` — replace placeholder IPs with real ones.

BUILD

Step 1 — Test connectivity

```
cd harrington-infrastructure-operations/ansible
ansible all -i inventory.ini -m ping
```

Expected output for each host:

```
HC-APP01 | SUCCESS => { "ping": "pong" }
HC-MON01 | SUCCESS => { "ping": "pong" }
```

If any host fails, fix SSH connectivity before proceeding.

Step 2 — Run the hardening playbook

```
ansible-playbook -i inventory.ini harden-linux.yml --diff
```

The `--diff` flag shows exactly what changes are being made to each file.

This will take approximately 3–5 minutes per host. Watch for: - `changed` — task made a change - `ok` — already in correct state - `failed` — error (see Troubleshooting)

Step 3 — Copy the PLAY RECAP

At the end of the playbook run you will see:

```
PLAY RECAP *****
HC-APP01 : ok=18  changed=12  unreachable=0  failed=0  skipped=2  ...
HC-MON01 : ok=18  changed=12  unreachable=0  failed=0  skipped=0  ...
```

Copy this entire PLAY RECAP — paste it into your submission.

Step 4 — Commit the playbook

```
git add ansible/
git commit -m "feat(ansible): Linux hardening playbook applied - Phase 3"
git push origin main
```

VERIFY

SSH into HC-APP01 and run each check:

Check 1 — SSH password auth disabled

```
sudo grep "PasswordAuthentication" /etc/ssh/sshd_config
```

Must show: PasswordAuthentication no

Check 2 — UFW active with correct rules

```
sudo ufw status verbose
```

Must show:

```
Status: active
To Action From
--
22/tcp ALLOW IN Anywhere
443/tcp ALLOW IN Anywhere
```

Take a screenshot of this output.

Check 3 — fail2ban running

```
sudo systemctl status fail2ban
sudo fail2ban-client status sshd
```

Must show: Active: active (running)

Take a screenshot of this output.

Check 4 — NTP synced

```
timedatectl status
ntpq -p
```

Must show: System clock synchronized: yes

Check 5 — Test SSH key auth (from your control machine)

```
ssh hcadmin@<HC-APP01-IP>
# Should connect without password prompt

# Attempt password login (should FAIL):
ssh -o PreferredAuthentications=password hcadmin@<HC-APP01-IP>
# Expected: Permission denied (publickey)
```

SUBMIT

Your Phase 3 submission must include:

#	Evidence	How to get it
1	Ansible PLAY RECAP	Copy from terminal after playbook run
2	<code>sudo ufw status verbose</code> screenshot	SSH into HC-APP01, run command, screenshot
3	<code>sudo systemctl status fail2ban</code> screenshot	SSH into HC-APP01, run command, screenshot
4	SSH password rejection output	<code>ssh -o PreferredAuthentications=password</code> showing <code>Permission denied</code>
5	GitHub commit URL	Commit with playbook run evidence

Anti-fake check: The PLAY RECAP must show `failed=0` for both hosts. The `changed` count must be greater than 0 (proves the playbook ran, not just reported).

Troubleshooting

Error	Fix
<code>Permission denied (publickey)</code> on ansible run	Run <code>ssh-copy-id hcadmin@<HOST-IP></code> first
<code>UNREACHABLE</code> for a host	Check VM is running: <code>ping <HOST-IP></code>
<code>FAILED: ufw enable</code>	Run <code>sudo apt install -y ufw</code> manually first
Locked out of SSH after hardening	Connect via VirtualBox console: <code>sudo ufw allow ssh</code>
<code>fail2ban</code> not starting	Check: <code>sudo journalctl -u fail2ban -n 20</code>